Frequency Dynamic Convolution for Dense Image Prediction

CVPR 2025
Chen et al.

Outline

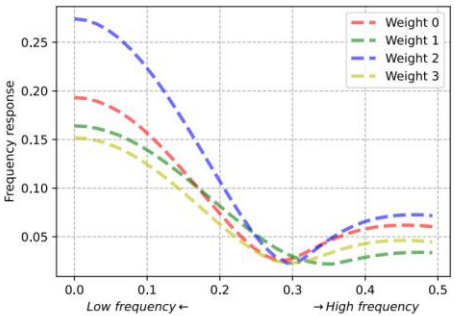
- Introduction
- Related Work
- Method
- Experiments
- Conclusion

Outline

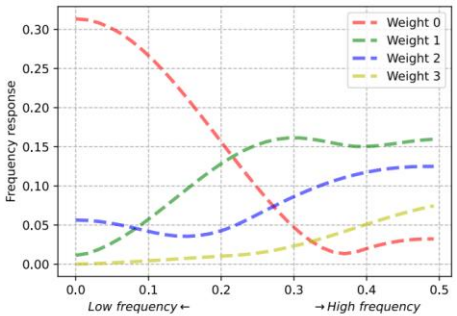
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Introduction

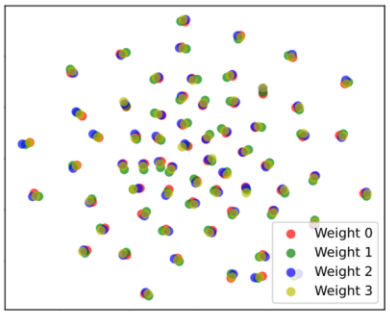
- Task
  - Convolution block of dense image prediction
- Problem
  - Previous dynamic convolutions lack of frequency diversity in their parallel weights.
  - Parallel weights of previous methods shows high similarity in parameter space.
  - Despite a significant increase in parameters, the model's ability improves little.
- Contribution
  - Propose FDConv, a novel dynamic convolution aim to enhance frequency adaptability.



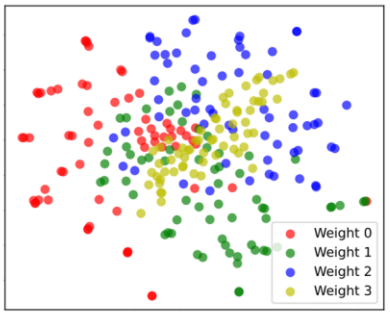
(a) Frequency response of ODConv



(b) Frequency response of FDConv



(c) t-SNE result of ODConv



(d) t-SNE result of FDConv

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Related Work

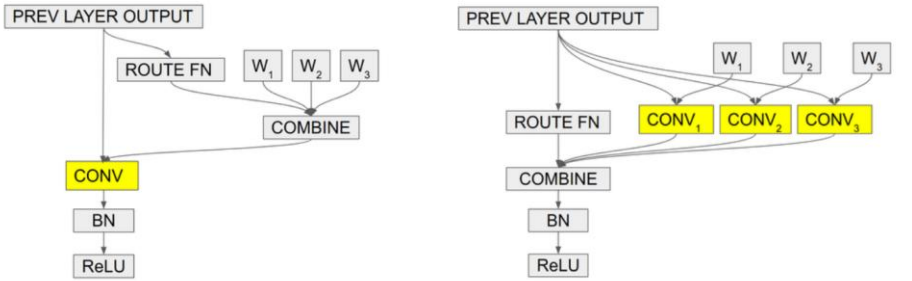
- CondConv: Conditionally Parameterized Convolutions for Efficient Inference, NeurIPS 2019, Yang et al.
- Omni-Dimensional Dynamic Convolution, ICLR 2022, Li et al.

Related Work

- CondConv: Conditionally Parameterized Convolutions for Efficient Inference, NeurIPS 2019, Yang et al.
- Omni-Dimensional Dynamic Convolution, ICLR 2022, Li et al.

Introduction (Yang et al.)

- Task
  - Convolution block of image classification / object detection
- Problems
  - In traditional CNN, the larger capacity means more layers or larger weight tensor, causing high computational cost during inference.
- Contributions
  - Propose CondConv, which compute convolutional kernels as a function of the input, leverage the capacity while maintaining efficient inference.



(a) CondConv: $(\alpha_1 W_1 + \ldots + \alpha_n W_n) * x$ (b) Mixture of Experts: $\alpha_1 (W_1 * x) + \ldots + \alpha_n (W_n * x)$

Method (Yang et al.)

- The final weight is linear combination of parallel weights and routing function:

$$Output(x) = \sigma((\alpha_1 \cdot W_1 + \ldots + \alpha_n \cdot W_n) * x) \qquad \alpha_i = r_i(x)$$

- The routing function is computed as linear of GAP of input followed by sigmoid:

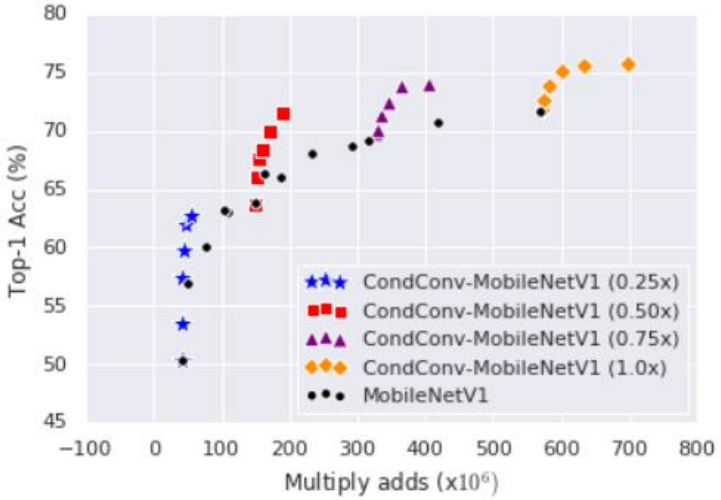
$$r(x) = \text{Sigmoid}(\text{GlobalAveragePool}(x)\ R)$$

- It is mathematically equivalent to the mixture of experts approach with soft routing function.

$$\sigma((\alpha_1 \cdot W_1 + \ldots + \alpha_n \cdot W_n) * x) = \sigma(\alpha_1 \cdot (W_1 * x) + \ldots + \alpha_n \cdot (W_n * x))$$

Experiments (Yang et al.)

- ImageNet Classification



	Baseline MADDs ($\times 10^6$)	Baseline Top-1 (%)	CondConv MADDs ($\times 10^6$)	CondConv Top-1 (%)
MobileNetV1 (1.0x)	567	71.9	600	73.7
MobileNetV2 (1.0x)	301	71.6	329	74.6
MnasNet-A1	312	74.9	325	76.2
ResNet-50	4093	77.7	4213	78.6
EfficientNet-B0	391	77.2	413	78.3

	Baseline MADDs (M)	Baseline Top-1 (%)	CondConv MADDs (M)	CondConv Top-1 (%)
MobileNetV1	571 (1.0x)	71.7 (1.0x)	190 (0.5x)	71.6 (0.5x)
MobileNetV2	585 (1.4x)	74.5 (1.4x)	329 (1.0x)	74.6 (1.0x)
MnasNet	340 (A2)	75.6 (A2)	329 (A1)	76.2 (A1)

Experiments (Yang et al.)

- COCO Object Detection

	Baseline		CondConv	
	MADDs ($\times 10^6$)	mAP	MADDs ($\times 10^6$)	mAP
MobileNetV1 (0.5x)	352	14.4	363	18.0
MobileNetV1 (0.75x)	730	18.2	755	21.0
MobileNetV1(1.0x)	1230	20.3	1280	22.4

Experiments (Yang et al.)

- Ablation study on Routing function

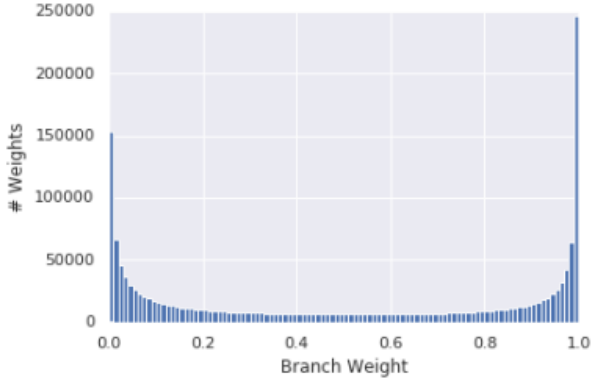
Routing Fn	MADDs ($\times 10^6$)	Valid Top-1 (%)
CC-MobileNetV1 (0.25x)	55.7	62.0
Single	55.5	56.5
Partially-shared	55.6	62.5
Hidden (small)	55.6	57.7
Hidden (medium)	55.9	62.2
Hidden (large)	57.8	54.1
Hierarchical	55.7	60.3
Softmax	55.7	60.5

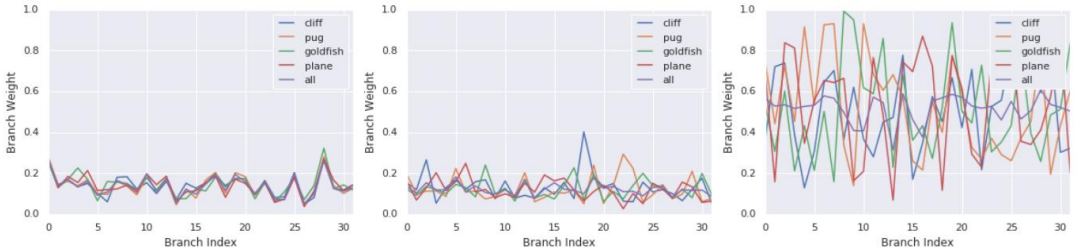
Experiments (Yang et al.)

- Ablation study on CondConv Layer Depth

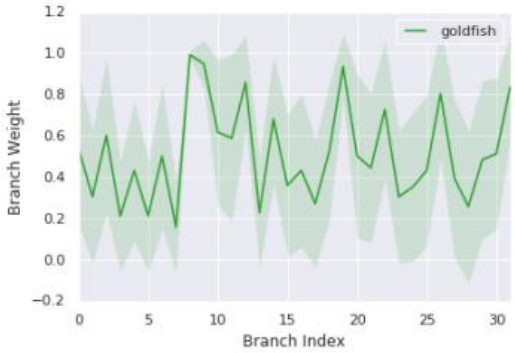
CondConv Begin Layer	MADDs ($\times 10^6$)	Valid Top-1 (%)
CC-MobileNetV1 (0.25x)	55.7	62.0
MobileNetV1 (0.25x)	41.2	50.0
1	56.3	62.5
5	56.0	62.0
7	55.7	62.0
13	52.5	59.5
15 (FC Only)	49.3	54.2
7 (No FC)	47.6	60.2

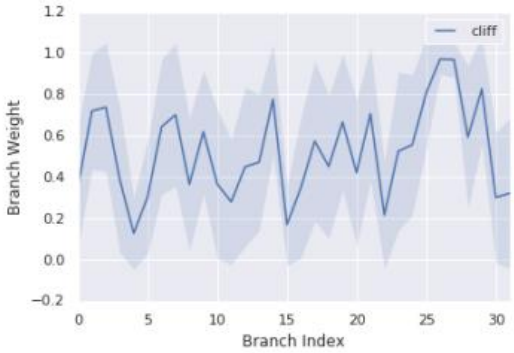
Analysis (Yang et al.)



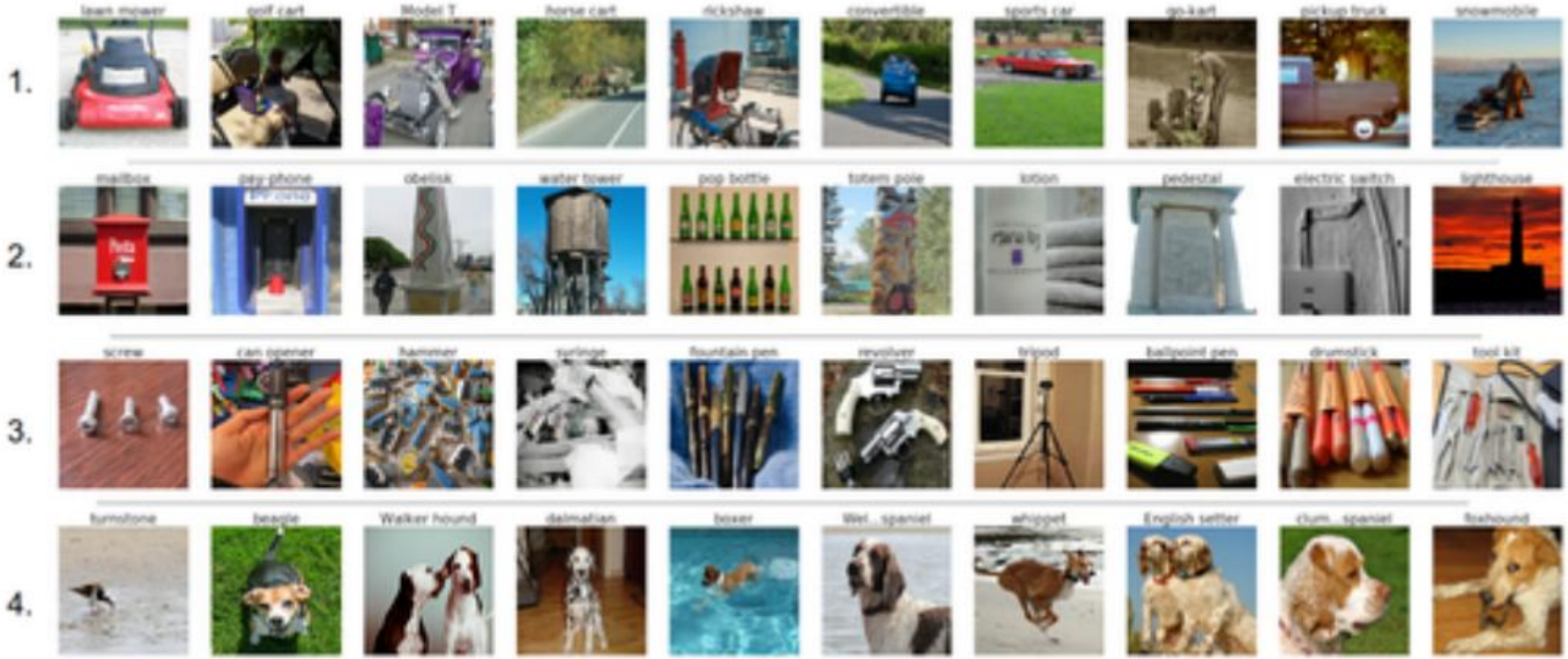


(a) Layer 12 (b) Layer 26 (c) Fully Connected (FC)





Analysis (Yang et al.)



Conclusion (Yang et al.)

- Propose CondConv, a new direction for increasing model capacity while maintaining efficient inference (dynamic convolution).

Related Work

- CondConv: Conditionally Parameterized Convolutions for Efficient Inference, NeurIPS 2019, Yang et al.
- Omni-Dimensional Dynamic Convolution, ICLR 2022, Li et al.

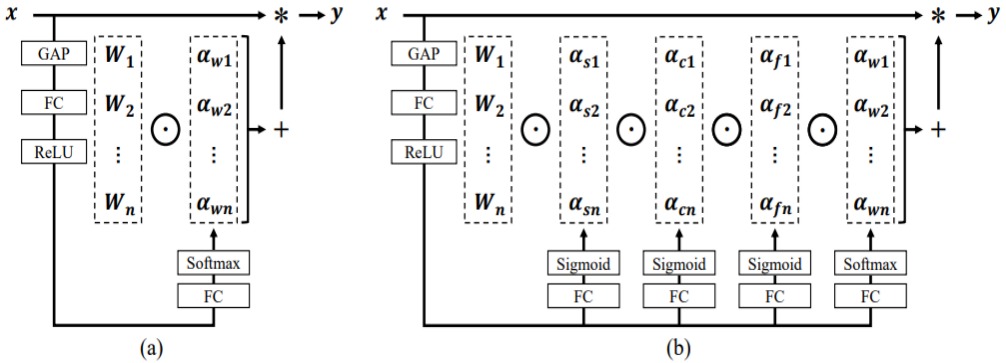
Introduction (Li et al.)

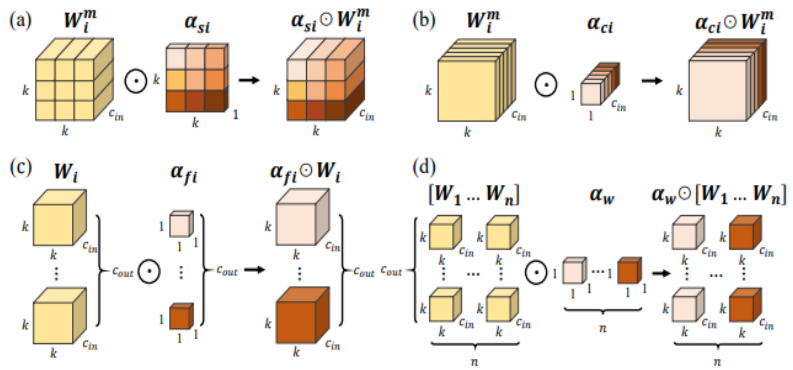
- Task
  - Convolution block of image classification / object detection
- Problem
  - Previous dynamic convolutions are only dynamic on dimension of kernel space.
  - Other dimension (spatial, in/out channel) are still static
  - Previous dynamic convolutions increase parameter size by n (# of weights) times.
- Contribution
  - Propose ODConv, a new dynamic convolution method utilitizing spatial and in/out channel dimension.

Method (Li et al.)

$$y = (\alpha_{w1} \odot \alpha_{f1} \odot \alpha_{c1} \odot \alpha_{s1} \odot W_1 + \ldots + \alpha_{wn} \odot \alpha_{fn} \odot \alpha_{cn} \odot \alpha_{sn} \odot W_n) * x$$

α_w :weight
α_f :output channel
α_c :input channel
α_s :spatial





Method (Li et al.)

- α_f α_c α_s are shared across weights
- Four share same $\mathrm{GAP} \rightarrow \mathrm{FC(r = 1/16)} \rightarrow \mathrm{ReLU}$ stem
- α_w use softmax, others use sigmoid
- use temperature annealing

$$\alpha_{wi} = \frac{\exp(z_i/\tau)}{\sum_{j=1}^{n} \exp(z_j/\tau)}$$

$$\alpha = \sigma\left(\frac{z}{\tau}\right) = \frac{1}{1 + \exp(-z/\tau)}$$

Experiments (Li et al.)

- Image Classification on ImageNet — MobileNets

Models	Params	MAdds	Top-1 Acc (%)	Top-5 Acc (%)
MobileNetV2 (1.0×)	3.50M	300.8M	71.65	90.22
+ CondConv (8×)	22.88M	318.1M	74.13 (↑2.48)	91.67 (↑1.45)
+ DyConv (4×)	12.40M	317.1M	74.94 (↑3.29)	91.83 (↑1.61)
+ DCD	5.72M	318.4M	74.18 (↑2.53)	91.72 (↑1.50)
+ ODConv (1×)	4.94M	311.8M	74.84 (↑3.19)	92.13 (↑1.91)
+ ODConv (4×)	11.52M	327.1M	**75.42** (↑**3.77**)	**92.18** (↑**1.96**)
MobileNetV2 (0.75×)	2.64M	209.1M	69.18	88.82
+ CondConv (8×)	17.51M	223.9M	71.79 (↑2.61)	90.17 (↑1.35)
+ DyConv (4×)	7.95M	220.1M	72.75 (↑3.57)	90.93 (↑2.11)
+ DCD	4.08M	222.9M	71.92 (↑2.74)	90.20 (↑1.38)
+ ODConv (1×)	3.51M	217.1M	72.43 (↑3.25)	90.82 (↑2.00)
+ ODConv (4×)	7.50M	226.3M	**73.81** (↑**4.63**)	**91.33** (↑**2.51**)
MobileNetV2 (0.5×)	2.00M	97.1M	64.30	85.21
+ CondConv (8×)	13.61M	110.0M	67.24 (↑2.94)	87.51 (↑2.30)
+ DyConv (4×)	4.57M	103.2M	69.05 (↑4.75)	88.37 (↑3.16)
+ DCD	3.06M	105.6M	69.32 (↑5.02)	88.44 (↑3.23)
+ ODConv (1×)	2.43M	101.8M	68.26 (↑3.9.6)	87.98 (↑2.77)
+ ODConv (4×)	4.44M	106.4M	**70.01** (↑**5.71**)	**89.01** (↑**3.80**)

Experiments (Li et al.)

- Image Classification on ImageNet — ResNets

Network	ResNet18				ResNet50			
Models	Params	MAdds	Top-1 Acc (%)	Top-5 Acc (%)	Params	MAdds	Top-1 Acc (%)	Top-5 Acc (%)
Baseline	11.69M	1.814G	70.25	89.38	25.56M	3.858G	76.23	93.01
+ CondConv (8×)	81.35M	1.894G	71.99 (↑1.74)	90.27 (↑0.89)	129.86M	3.978G	76.70 (↑0.47)	93.12 (↑0.11)
+ DyConv (4×)	45.47M	1.861G	72.76 (↑2.51)	90.79 (↑1.41)	100.88M	3.965G	76.82 (↑0.59)	93.16 (↑0.15)
+ DCD	14.70M	1.841G	72.33 (↑2.08)	90.65 (↑1.27)	29.84M	3.944G	76.92 (↑0.69)	93.46 (↑0.45)
+ ODConv (1×)	11.94M	1.838G	73.10 (↑2.85)	91.10 (↑1.72)	28.64M	3.916G	77.96 (↑1.73)	93.84 (↑0.83)
+ ODConv (4×)	44.90M	1.916G	**73.97** (↑**3.72**)	**91.35** (↑**1.97**)	90.67M	4.080G	**78.52** (↑**2.29**)	**94.01** (↑**1.00**)
+ SE	11.78M	1.816G	70.98 (↑0.73)	90.03 (↑0.65)	28.07M	3.872G	77.31 (↑1.08)	93.63 (↑0.62)
+ CBAM	11.78M	1.818G	71.01 (↑0.76)	89.85 (↑0.47)	28.07M	3.886G	77.46 (↑1.23)	93.59 (↑0.58)
+ ECA	11.69M	1.816G	70.60 (↑0.35)	89.68 (↑0.30)	25.56M	3.870G	77.34 (↑1.11)	93.64 (↑0.63)
+ CGC	11.69M	1.827G	71.60 (↑1.35)	90.35 (↑0.97)	25.59M	3.877G	76.79 (↑0.56)	93.37 (↑0.36)
+ WeightNet	11.93M	1.826G	71.56 (↑1.31)	90.38 (↑1.00)	30.38M	3.885G	77.51 (↑1.28)	93.69 (↑0.68)
+ WE (*)	11.90M	1.820G	71.00 (↑0.75)	90.00 (↑0.62)	28.10M	3.860G	77.10 (↑0.87)	93.50 (↑0.49)

Res101

Models	Params	MAdds	Top-1 Acc (%)	Top-5 Acc (%)
ResNet101	44.55M	7.570G	77.41	93.67
+ SE	49.29M	7.593G	78.42 (↑1.01)	94.15 (↑0.48)
+ CBAM	49.30M	7.617G	78.50 (↑1.09)	94.20 (↑0.53)
+ ECA	44.55M	7.590G	78.60 (↑1.19)	94.34 (↑0.67)
+ ODConv (1×)	50.82M	7.675G	78.98 (↑1.57)	94.38 (↑0.71)
+ ODConv (2×)	90.44M	7.802G	**79.27** (↑**1.86**)	**94.47** (↑**0.80**)

Experiments (Li et al.)

- Object Detection on COCO, backbones are pretrained by ImageNet classification

Backbone Models	Detectors	$AP(\%)$	$AP_{50}(\%)$	$AP_{75}(\%)$	$AP_S(\%)$	$AP_M(\%)$	$AP_L(\%)$	Params	MAdds
ResNet50	Faster R-CNN	37.2	57.8	40.4	21.5	40.6	48.0	43.80M (23.51M)	207.07G (76.50G)
+ CondConv (8×)		38.1	58.9	41.5	22.4	42.1	48.7	133.75M (113.46M)	207.08G (76.51G)
+ DyConv (4×)		38.3	59.7	41.6	22.6	42.3	49.4	119.12M (98.83M)	207.23G (76.66G)
+ DCD		38.1	59.3	41.3	21.9	42.0	49.5	48.08M (27.79M)	207.20G (76.63G)
+ ODConv (1×)		39.0	60.5	42.3	**23.4**	42.3	50.5	46.88M (26.59M)	207.18G (76.61G)
+ ODConv (4×)		**39.2**	**60.7**	**42.6**	23.1	**42.6**	**51.0**	108.91M (88.62M)	207.42G (76.85G)
MobileNetV2 (1.0×)		31.3	51.1	33.1	17.4	33.5	41.2	21.13M (2.22M)	122.58G (24.45G)
+ CondConv (8×)		33.7	54.9	35.6	19.3	36.4	43.7	31.54M (12.63M)	122.59G (24.46G)
+ DyConv (4×)		34.5	55.6	36.5	19.8	37.3	44.7	30.02M (11.12M)	123.01G (24.88G)
+ DCD		33.3	53.0	35.1	19.9	36.1	43.2	23.34M (4.44M)	123.01G (24.88G)
+ ODConv (1×)		34.3	55.6	36.5	**20.7**	37.3	44.5	22.56M (3.66M)	123.00G (24.87G)
+ ODConv (4×)		**35.1**	**56.7**	**37.0**	20.6	**38.0**	**45.2**	29.14M (10.24M)	123.02G (24.89G)
ResNet50	Mask R-CNN	38.0	58.6	41.5	21.6	41.5	49.2	46.45M (23.51M)	260.14G (76.50G)
+ CondConv (8×)		38.8	59.3	42.3	22.5	42.5	50.3	136.4M (113.46M)	260.15G (76.51G)
+ DyConv (4×)		39.2	60.3	42.5	23.0	42.9	51.4	121.77M (98.83M)	260.30G (76.66G)
+ DCD		38.8	59.8	42.2	23.1	42.7	49.8	50.73M (27.79M)	260.27G (76.63G)
+ ODConv (1×)		39.9	61.2	43.5	23.6	**43.8**	**52.3**	49.53M (26.59M)	260.25G (76.61G)
+ ODConv (4×)		**40.1**	**61.5**	**43.6**	**24.0**	43.6	**52.3**	111.56M (88.62M)	260.49G (76.85G)
MobileNetV2 (1.0×)		32.2	52.1	34.2	18.4	34.4	42.4	23.78M (2.22M)	175.66G (24.45G)
+ CondConv (8×)		34.4	55.4	36.6	19.8	36.9	44.6	34.19M (12.63M)	175.67G (24.46G)
+ DyConv (4×)		35.2	56.2	37.5	**20.7**	38.0	45.5	32.68M (11.12M)	176.09G (24.88G)
+ DCD		34.3	54.9	36.6	20.6	37.1	44.8	26.00M (4.44M)	176.09G (24.88G)
+ ODConv (1×)		35.0	56.1	37.3	19.9	37.7	**46.2**	25.22M (3.66M)	176.08G (24.87G)
+ ODConv (4×)		**35.8**	**57.0**	**38.1**	20.5	**38.5**	45.9	31.80M (10.24M)	176.10G (24.89G)

Experiments (Li et al.)

- Ablation study

Models	r	Params	MAdds	Top-1 Acc (%)	Top-5 Acc (%)
ResNet18	-	11.69M	1.814G	70.25	89.38
+ ODConv ($1\times$)	1/4	12.58M	1.839G	**73.41**	**91.29**
	1/8	12.15M	1.838G	73.11	91.10
	1/16	11.94M	1.838G	73.10	91.10

Models	n	Params	MAdds	Top-1 Acc (%)	Top-5 Acc (%)
ResNet18	-	11.69M	1.814G	70.25	89.38
+ ODConv ($r = 1/16$)	$1\times$	11.94M	1.838G	73.10	91.10
	$2\times$	22.93M	1.872G	73.59	91.08
	$3\times$	33.92M	1.894G	73.77	91.35
	$4\times$	44.90M	1.916G	73.97	91.35
	$8\times$	88.84M	2.006G	**74.08**	**91.44**

Experiments (Li et al.)

- Ablation study

Models	α_{si}	α_{ci}	α_{fi}	α_{wi}	Params	MAdds	Top-1 Acc (%)	Top-5 Acc (%)
ResNet18	-	-	-	-	11.69M	1.814G	70.25	89.38
+ ODConv ($r = 1/4$)	✓	-	-	-	11.98M	1.827G	72.42	90.76
	-	✓	-	-	12.26M	1.827G	72.07	90.59
	-	-	✓	-	12.28M	1.827G	71.46	90.43
	✓	✓	-	-	12.27M	1.827G	73.13	91.14
	✓	-	✓	-	12.29M	1.827G	72.80	90.99
	-	✓	✓	-	12.57M	1.829G	72.20	90.67
	✓	✓	✓	-	12.58M	1.839G	73.41	91.29
	✓	✓	✓	✓	45.54M	1.953G	**74.33**	**91.53**

ResNet50	Speed on GPU	Speed on CPU
Baseline model	652.2	6.4
+ CondConv	425.1	4.0
+ DyConv	326.6	3.8
+ DCD	400.0	3.6
+ ODConv (1×)	293.9	3.9
+ ODConv (4×)	152.7	2.5

MobileNetV2 (1.0×)	Speed on GPU	Speed on CPU
Baseline model	1452.3	17.3
+ CondConv	1076.9	14.8
+ DyConv	918.2	11.9
+ DCD	875.7	9.5
+ ODConv (1×)	1029.2	12.8
+ ODConv (4×)	608.3	11.2

Conclusion (Li et al.)

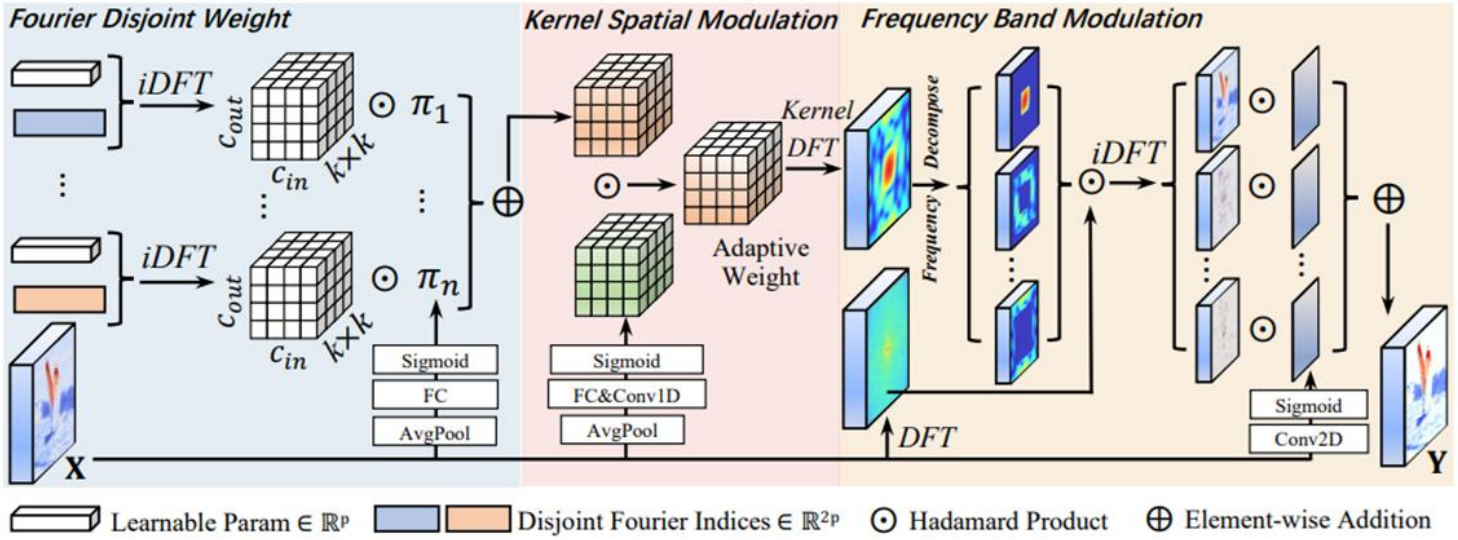
- Propose ODConv, a new dynamic convolution method utilitizing spatial and in/out channel dimension.
- Spatial attention matters more than combination of matrix and channel attention.
- ODConv can surpass previous method with only one weight.

Outline

- Introduction
- Related Work
- Method
- Experiments
- Conclusion

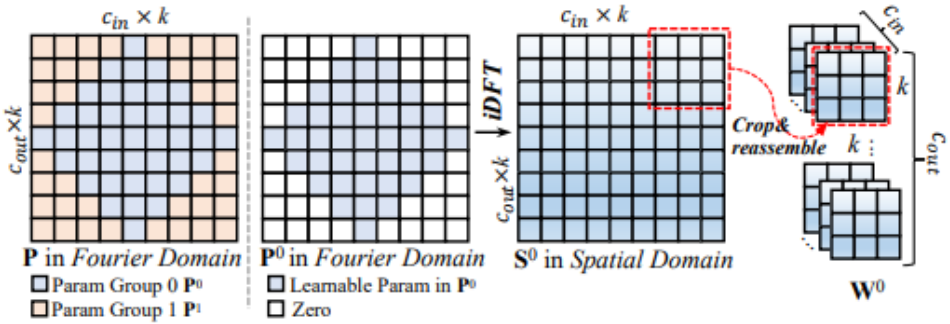
Method

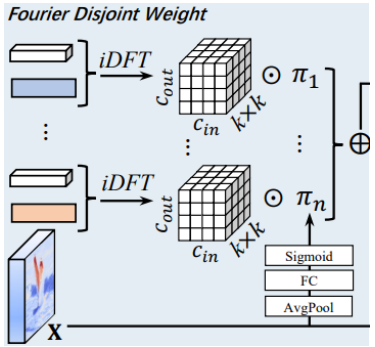
- Overview
  - FDW followed by KSM followed by FBM



Method

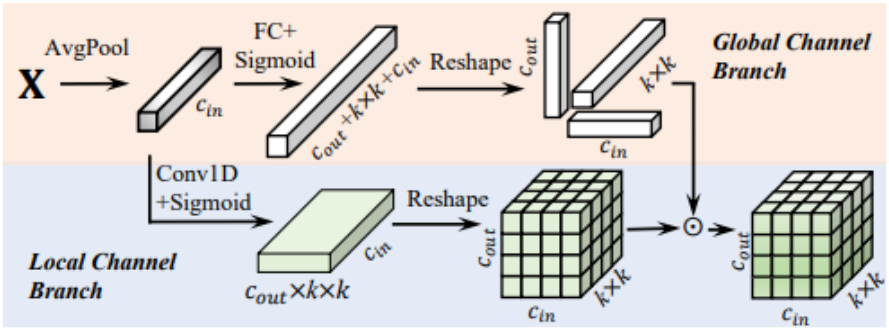
- Fourier Disjoint Weight (FDW)
  - Total number of params is set to $kC_in \times kC_out$, representing spectral map of weights.
  - Divide params into n groups by L_2 norm to spectral center.
  - For i-th group, apply IDFT and reshape it to $k \times k \times C_{in} \times C_{out}$ to form W_i.
  - Form dynamic weight as linear combination of W_i and routing function π_i.

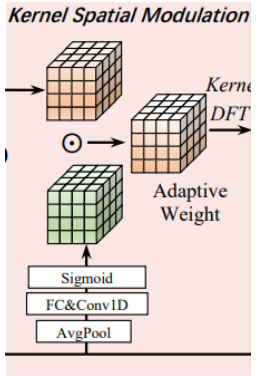




Method

- Kernel Spatial Modulation (KSM)
  - AP X and do Conv1D+sigmoid and reshape them to form local branch.
  - AP X and do FC+sigmoid to form three 1D vectors, outer-product them to form global branch.
  - Do elementwise multiplication of two branch to form spatial-channel attention mask.





Method

- Frequency Band Modulation (FBM)
  - For each $k \times k$ kernels W of each input/output channels, do:
  - Do Fourier transform to get $\mathcal{F}(W)$.
  - Divide $\mathcal{F}(W)$ into B groups by L_∞ norm in spectral map.
  - Do inverse Fourier transform for each group to form W_b

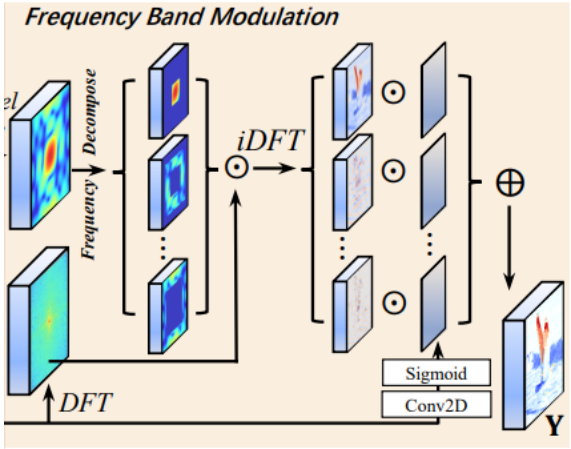
$$\mathbf{W}_b = \mathcal{F}^{-1}(\mathcal{M}_b \odot \mathcal{F}(\mathbf{W}))$$

  - Convolve W_b with X to get Y_b.

$$\mathbf{Y}_b = \mathbf{W}_b * \mathbf{X}$$

  - Apply spatial modulation mask A_b to Y_b and sum up to get Y.

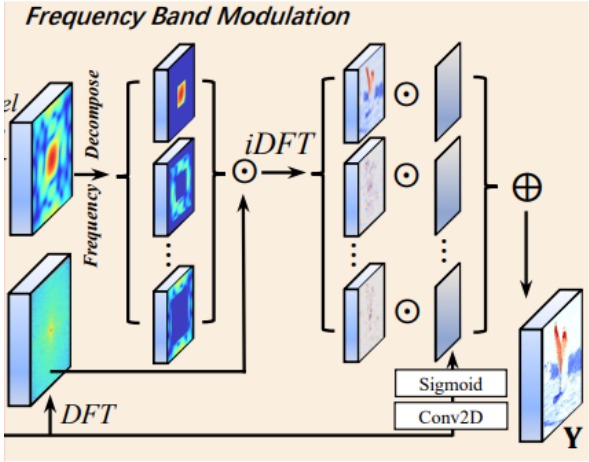
$$\mathbf{Y} = \sum_{b=0}^{B-1} (\mathbf{A}_b \odot \mathbf{Y}_b)$$



Method

- Frequency Band Modulation (FBM)
  - The process can also be rewritten as:

$$\mathbf{Y}_b = \mathcal{F}^{-1}\left((\mathcal{M}_b \odot \mathcal{F}(\mathbf{W})) \odot \mathcal{F}(\mathbf{X})\right)$$



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Experiments

- Object Detection and segmentation on COCO with ResNet50

Models	Params	FLOPs	AP^{box}	AP^{mask}
Faster R-CNN	$43.80_{(23.5)}$M	207.1G	37.2	
+ CondConv[NIPS2019] (8×) [73]	+90.0M	+0.01G	38.1	-
+ DY-Conv[ICLR2022] (4×) [63]	+75.3M	+0.16G	38.3	-
+ DCD[ICLR2021] [37]	+4.3M	+0.13G	38.1	
+ ODConv[ICLR2022] (4×) [32]	+65.1M	+0.35G	39.2	-
+ FDConv (Ours)	+3.6M	+1.8G	**39.4**	
Mask R-CNN	$46.5_{(23.5)}$M	260.1	39.6	36.4
+ DY-Conv[CVPR2020] (4×) [63]	+75.3M	+0.16G	39.6	36.6
+ ODConv[ICLR2022] (4×) [32]	+65.1M	+0.35G	42.1	38.6
+ KW[ICML2024] (1×) [31]	+2.5M	-	41.8	38.4
+ KW[ICML2024] (4×) [31]	+76.5M	-	42.4	**38.9**
+ FDConv (Ours)	+3.6M	+1.8G	**42.4**	38.6

Experiments

- Object Detection and segmentation on COCO with Mask R-CNN

Model	Params	FLOPs	AP^{box}	AP^{mask}
ConvNeXt-T [47]	48M	262G	43.4	39.7
+ KW [ICML2024] [31]	52M	262G	44.8	40.6
+ FDConv (Ours)	51M	263G	**45.2**	**40.8**
Swin-T[46]	48M	267G	42.7	39.3
+ FDConv (Ours)	51M	268G	**44.5**	**40.5**

Experiments

- ADE20K semantic segmentation with UPerNet

Method	Params	FLOPS	mIoU	
			SS	MS
ResNet-50 [23]	66M	947G	40.7	41.8
ResNet-101 [23]	85M	1029G	42.9	44.0
R50 + PEDConv[BMVC2021] [24]	72M	947G	42.8	43.9
R50 + ODConv[ICLR2022](4×) [32]	131M	947G	43.3	44.4
R50 + KW[ICML2024](4×) [31]	141M	947G	43.5	44.6
R50 + FDConv (Ours)	70M	949G	**43.8**	**44.9**

Experiments

- Cityscapes semantic segmentation with Mask2Former

Model	Backbone	mIoU
Mask2Former[CVPR2022] [8]	ResNet-50	79.4
+ FDConv (Ours)	ResNet-50	**80.4 (+1.0)**

Experiments

- ADE20K semantic segmentation with Mask2Former and MaskDINO
    - backbones are pretrained with ImageNet-22K

Model	Backbone	mIoU
Mask2Former[CVPR2022] [8]	Swin-B†	53.9
+ FDConv (Ours)	Swin-B†	**54.9 (+1.0)**
MaskDINO[CVPR2023] [34]	Swin-L†	56.6
+ FDConv (Ours)	Swin-L†	**57.2 (+0.5)**

Experiments

- Ablation studies

Models	Params	AP^{box}
Mask R-CNN	$46.5_{(23.5)}$M	39.6
+ FDW	+ 1.3M	40.6 (+1.0)
+ FDW + KSM (local only)	+1.6M	41.2 (+1.6)
+ FDW + KSM	+ 3.5M	41.8 (+2.2)
+ FDW + KSM + FBM	+ 3.6M	**42.4 (+2.8)**

Number of Weights (n)	$n = 4$	$n = 16$	$n = 64$	$n = 256$
AP^{box}	42.0	42.2	**42.4**	**42.4**

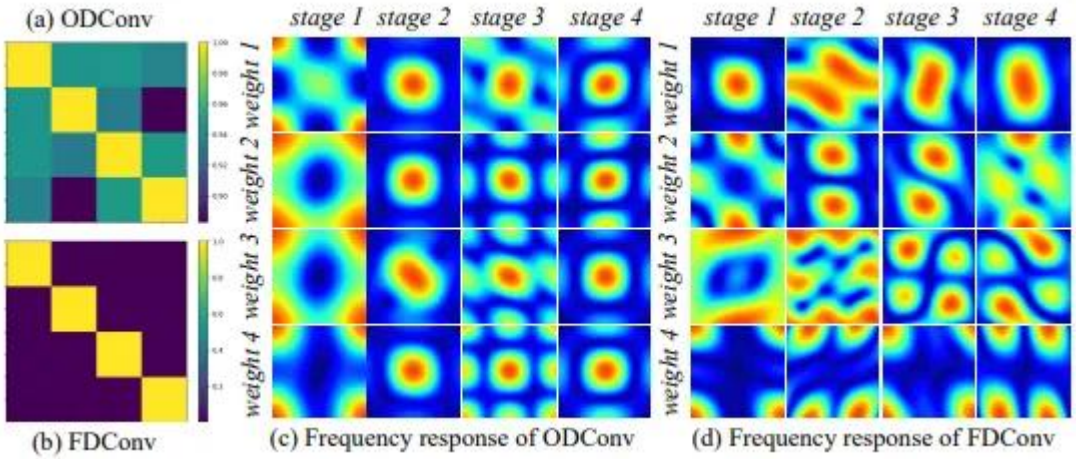
Model	Standard Conv	ODConv [7]	FDConv (Ours)
FPS ($n = 1$)	736.1	-	-
FPS ($n = 4$)		**498.7**	454.8
FPS ($n = 16$)	-	325.3	**450.2**
FPS ($n = 64$)	-	135.7	**447.1**

Experiments

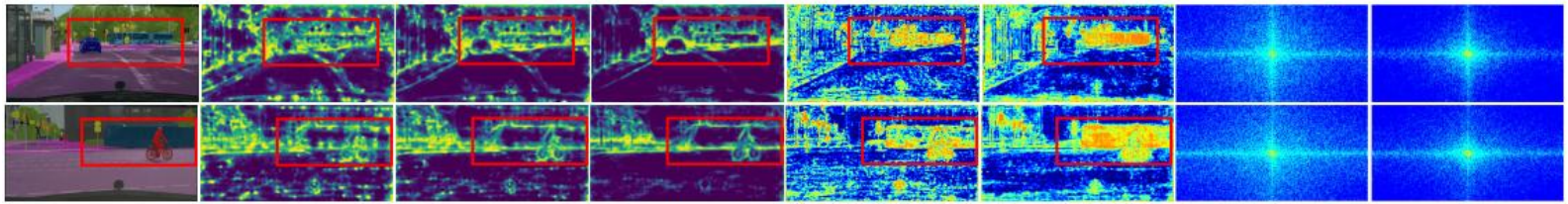
- Ablation studies

Frequency Band	Set lowest band to 1.0	AP^{box}
2: $[0, \frac{1}{4}), [\frac{1}{4}, \frac{1}{2}]$	✓	42.2
3: $[0, \frac{1}{8}), [\frac{1}{8}, \frac{1}{4}), [\frac{1}{4}, \frac{1}{2}]$	✓	42.3
4: $[0, \frac{1}{16}), [\frac{1}{16}, \frac{1}{8}), [\frac{1}{8}, \frac{1}{4}), [\frac{1}{4}, \frac{1}{2}]$	✓	**42.4**
5: $[0, \frac{1}{32}), [\frac{1}{32}, \frac{1}{16}), [\frac{1}{16}, \frac{1}{8}), [\frac{1}{8}, \frac{1}{4}), [\frac{1}{4}, \frac{1}{2}]$	✓	42.2
4: $[0, \frac{1}{16}), [\frac{1}{16}, \frac{1}{8}), [\frac{1}{8}, \frac{1}{4}), [\frac{1}{4}, \frac{1}{2}]$		42.1

Analyses


(a) ODConv
(b) FDConv
stage 1
stage 2
stage 3
stage 4
weight 1
weight 2
weight 3
weight 4
(c) Frequency response of ODConv
(d) Frequency response of FDConv

Analyses



(a) Input image & GT (b) Modulation for $[\frac{1}{16}, \frac{1}{8})$ (c) Modulation for $[\frac{1}{8}, \frac{1}{4})$ (d) Modulation for $[\frac{1}{4}, \frac{1}{2}]$ (e) Feature before FBM (f) Feature after FBM (g) Spectrum before FBM (h) Spectrum after FBM

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Conclusion

- Parallel kernels in existing methods have highly similar frequency responses.
- Propose FDConv, learning frequency-disjoint weight and modulates frequency bands spatially.
- The total number of parameter each weight is restricted to k×k×Cin×Cout.